Duke University **April 2020**
Math 106L: Laboratory Calculus and Functions II

End of Term Assignment 2

- This is an assignment with exam-style questions.
- There are 60 points available in total.
- You should use a spreadsheet on questions where its use is indicated.
- When you have completed the assignment, you should scan it in and upload it to Gradescope.
- All work done should be your own, completed individually with no assistance from anyone else.
- Sign the honor code below to indicate that you agree to this.

Name: _____

Honor Code:

I have adhered to the Duke Community Standard in completing this exam.

Signature: _____

1. (12 points) Consider the differential equation $\frac{dy}{dt} = 4t + 7$.

   (a) Show that $f(t) = 2t^2 + 7t$ and $g(t) = 2t^2 + 7t - 5$ are both solutions to the above differential equation.

   (b) Solve the initial value problem with $y(1) = 4$.

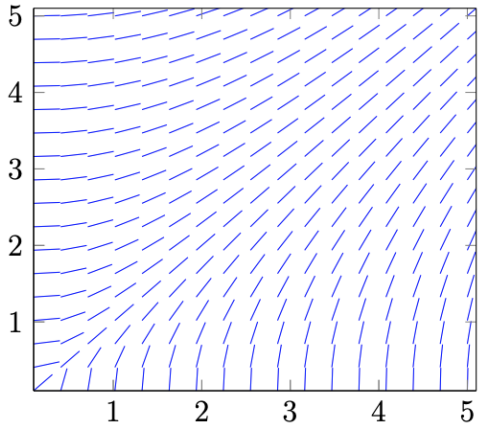
   (c) What happens to any solution of this differential equation as $t \to \infty$?

2. (8 points) Consider the differential equation $\frac{dy}{dx} = xy - y$, and suppose $y = f(x)$ is a solution.

   (a) Is $y = 3f(x)$ a solution? Justify.

   (b) Is $y = f(x) + 3$ a solution? Justify.

3. (16 points) The slope field for $\frac{dy}{dx} = \frac{x}{y}$ is provided below.



(a) On the graph above, sketch solutions for the initial conditions $y(0) = 2$, $y(3) = 1$, and $y(1) = 1$.

(b) Using the provided slope field, find the solution (for $x > 0$ and $y > 0$) to the initial value problem $\frac{dy}{dx} = \frac{x}{y}$ with initial condition $y(2) = 2$.

(c) Sketch the slope field for Quadrant II. Also sketch the solution with initial condition $y(0) = 2$ in Quadrant II.

4. (12 points) Consider the differential equation $\frac{dy}{dt} = \frac{f(y)}{g(t)}$, where $f(y)$ is some function of y and $g(t)$ is some function of t.

   (a) Explain why this differential equation is separable.

   (b) Let $f(y) = y^2$ and $g(t) = t$. Find the solution to the differential equation with initial value $y(1) = 6$.

5. (12 points) Consider the differential equation $\frac{dy}{dx} = x^4(y^2 - y - 2)$. Find all equilibrium solutions and determine their stability.